ECO101
Practice Sheet 1 - Production Possibilities Frontier & Gains from Trade

1. The table below displays the maximum daily production of vanilla and chocolate ice creams (in liters) among the top ice-cream producing companies in Bangladesh:

Company	Vanilla Ice Cream	Chocolate Ice Cream
Delight	60	30
Frosty	40	90
Snowy	70	50

a. What is the opportunity cost of Delight producing 10 liters of chocolate ice cream per day?

b. Which company has the absolute advantage in producing vanilla ice cream?

c. Which company has the comparative advantage in producing vanilla ice cream?

d. Which company has the comparative advantage in producing chocolate ice cream?

2. Suppose we have two countries, Country A and Country B, both producing combinations of clothes and food. Their production possibilities are represented by points along their respective Production Possibility Frontiers (PPFs).

Country A: Moving from Point E: (Food = 25, Clothes = 75)

To Point F: (Food = 55, Clothes = 45)

Country B: Moving from Point E: (Food = 35, Clothes = 65)

To Point F: (Food = 65, Clothes = 40)

a. Determine which country has the comparative advantage in Food and which in Clothes.

3. Consider a simple economy producing two goods: cars and milk. The following table gives several points on this economy's production possibility frontier.

a. Graph the economy's production possibility frontier.

Points	Cars (hundreds per year)	Milk (1000's of gallon per year)
A	0	60
B	1	50
C	2	30
D	3	0

b. After drawing the PPF locate a point which will be unattainable for this economy.
c. Calculate the opportunity cost of producing cars.

4. Anna and Leo run juice bars producing **juices and wraps**.

Anna's Production table

Item	Minutes to produce 1	Quantity per hour
Juice	3	20
Wrap	3	20

Leo's Production table

Item	Minutes to produce 1	Quantity per hour
Juice	12	5
Wrap	2	30

a. Calculate Anna's opportunity cost of producing a wrap .

b. Calculate Leo's opportunity cost of producing a wrap.

c. Who has a comparative advantage in producing juice?

d. Initially both Anna and Leo split their time equally between juices and wraps:

- Anna produces **10 juices and 10 wraps**
- Leo produces **2.5 juices and 15 wraps**

And they agree to trade **6 juices for 12 wraps.**

Calculate the final amount of juices and wraps each person has after the trade, and explain who gains and how much in each product compared to their original output.

5. It takes Maria 4 hours to make a shirt and 2 hours to make a scarf. It takes John 3 hours to make a shirt and 1 hour to make a scarf.

a. Who has the absolute advantage in making a scarf
b. Who has the absolute advantage in making a shirt
c. Who has the comparative advantage in each good?